

Enhancing ASD Diagnosis with Contrastive and Non-Contrastive Models from Neuroimaging Data

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Abstract—Autism Spectrum Disorder (ASD) is a neurodevelopmental condition which impacts social interaction, communication, and behavior. This study introduces a self-supervised learning approach for ASD prediction using resting-state functional MRI (rs-fMRI) scans from the ABIDE I dataset, employing contrastive and non-contrastive models. The contrastive model enhances learning by maximizing agreement between augmented views of the same scan, while the non-contrastive model reconstructs input data through an autoencoder. Both models, pre-trained on ImageNet, were fine-tuned for ASD classification using Resnet18. Our results show that the contrastive model achieved 97.6% accuracy, with a precision of 95.7%, recall of 99.2%, and F1 score of 97.4%. The non-contrastive model reached 95.4% accuracy, with precision, recall, and F1 scores of 99.3%, 96.46%, and 97.82%. These findings highlight the performance of self-supervised learning in ASD diagnosis, showcasing robust pattern-capturing capabilities for neuroimaging data.

Index Terms—Autism Spectrum Disorder, SimCLR, SwAV, Self-Supervised Learning

I. INTRODUCTION

Autism Spectrum Disorder (ASD) is a complex neurodevelopmental condition marked by difficulties in social interaction, communication, and repetitive behaviors [1]. Early diagnosis and intervention are crucial for improving outcomes in individuals with ASD, underscoring the need for accurate and efficient diagnostic tools [2]. Recent advances in machine learning have shown significant promise in assisting ASD diagnosis through the analysis of neuroimaging data [3]. Among neuroimaging modalities, resting-state functional MRI (rs-fMRI) is widely used to capture intrinsic brain activity without task-based stimuli, providing valuable insights into brain functional connectivity and organization. This modality is particularly promising for understanding ASD's underlying mechanisms [4]. However, extracting meaningful information from rs-fMRI data remains challenging due to the data's high dimensionality and complex spatial-temporal patterns. Prior work has applied various machine learning techniques to rs-fMRI data, but limited interpretability and high computational demands have posed constraints.

To address these challenges, we explore self-supervised learning techniques for ASD prediction using rs-fMRI scans. Self-supervised learning has gained traction as a powerful method for representation learning from unlabeled data. By capturing inherent structures within the data, self-supervised models can learn discriminative features without manual annotations, supporting the development of robust, scalable diagnostic models. Our study investigates two types of self-supervised approaches: contrastive feature learning and non-contrastive feature learning. The contrastive learning model in this study leverages the SimCLR (Simple Contrastive Learning

of Representations) framework, which maximizes agreement between augmented views of the same rs-fMRI scan while minimizing agreement between views from different scans [5]. The non-contrastive approach utilizes the SwAV (Swapped Augmented Vectors) model, which employs an autoencoder to capture essential patterns and variations in rs-fMRI data relevant for ASD prediction. Both models were trained on the ABIDE I dataset, a comprehensive dataset of rs-fMRI scans from individuals with ASD and typically developing individuals, enabling a robust comparison of these self-supervised methods.

Our contributions are as follows:

- **Integration of Self-Supervised Learning:** This research applies self-supervised learning techniques to ASD diagnosis using rs-fMRI scans, offering a new approach for feature extraction in neuroimaging data.
- **Comparative Analysis of Feature Models:** We provide a detailed comparative analysis of contrastive and non-contrastive learning approaches for ASD diagnosis, demonstrating each method's strengths.
- **Enhanced ASD Diagnostic Performance:** Extensive experiments employing epoch-freezing fine-tuning techniques resulted in improved feature learning, enhancing overall diagnostic accuracy and performance.

The paper is structured as follows: Section 2 reviews related work; Section 3 details the proposed methodology, including preliminaries and proposed approach; Section 4 presents results and discussion; and Section 5 provides conclusions and future directions.

II. RELATED WORK

Various machine learning (ML) approaches have been explored for autism spectrum disorder (ASD) classification. M. Duda et al. [6] applied machine learning algorithms such as Random Forest (RF), Decision Trees (DT), Support Vector Machines (SVM), Naïve Bayes, and log-linear classifiers to classify ASD. Their results were satisfactory, with the accuracy assessed through an independent sample of children with and without autism. While effective, these methods require extensive feature engineering, and the interpretability of the models can be limited. Wall et al. [7] focused on simplifying autism diagnosis by reducing the set of questions needed for assessment. They developed a data-driven approach using the ADTree classifier, which reduced the number of questions by 72.4% compared to the original ADOS questionnaire. This method showed a significant advantage in reducing the diagnostic burden, though limitations include potential information loss due to question reduction.

Other studies by Wall et al. [7], Thabtah et al. [8], Duda et al. [6], and Karampasi et al. [9] highlight the potential of automated ML algorithms for faster diagnosis of ASD. Feature selection methods, such as ranking based on mutual information, improved classification accuracy. However, these approaches often rely on structured data and may struggle with high-dimensional neuroimaging data due to computational demands. Shahid et al. [10] explored the use of artificial neural networks (ANNs) for healthcare decision-making, showcasing their potential for rapid and real-time outputs. Their findings also emphasized that neural networks outperform traditional ML algorithms in classification tasks. However, neural networks are often computationally intensive and require large datasets for effective training, which may not always be available for ASD diagnosis.

Neuroimaging-based studies, such as those by Shen et al. [11] and Kasari et al. [12], utilized advanced technological methods to identify essential biomarkers for ASD. These methods have contributed to the development of diagnostic databases and age-specific treatment recommendations, enhancing the clinical utility of neuroimaging. Nevertheless, neuroimaging data is high-dimensional and can be challenging to interpret without advanced feature extraction techniques. Retico et al. [13] applied ML to structural MRI (sMRI) for Alzheimer's disease prediction, demonstrating how ML can help identify brain regions of interest. While the study focused on Alzheimer's, its approach underscores the potential of ML in neuroimaging-based classification problems, applicable to ASD diagnosis. Ghazal et al. [14] employed a transfer learning model using the AlexNet architecture to classify autism-related facial features, achieving a validation accuracy of 87.7%. This method is beneficial for early ASD prediction, yet facial feature-based approaches may be limited in generalizability to neuroimaging data. Other works include multiple classifications of autism using machine learning techniques [15], furthermore recent advanced methods which include federated learning is also evaluated for autism detection [16].

This paper builds on these efforts by implementing self-supervised learning models for ASD prediction using rs-fMRI data. Unlike traditional transfer learning approaches, our models leverage unlabeled data through self-supervised techniques, offering a scalable solution for high-dimensional brain scans. Our approach aims to address limitations of previous studies by capturing meaningful patterns from neuroimaging data, enhancing model interpretability, and improving diagnostic accuracy for ASD.

III. PROPOSED METHODOLOGY

In this section, we present the fine-tuned self-supervised learning framework developed for ASD prediction, detailing the contrastive and non-contrastive approaches, data preprocessing steps, and model architectures used to effectively capture diagnostic features from rs-fMRI data. Fig 1 depicts the overall model pipeline used in this study.

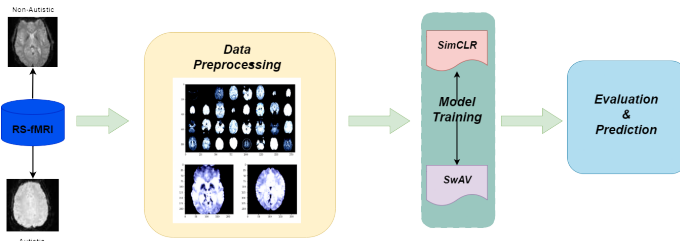


Fig. 1: Flowchart depicting the model pipeline.

A. Self Supervised Learning

By leveraging the vast amount of available unlabeled rs-fMRI data, self-supervised learning models can extract relevant information and build representations that are specific to individuals' neuroimaging data. Fig 2 shows the basic representation of self-supervised learning architecture. In the context of autism spectrum disorder (ASD) diagnosis using resting-state functional MRI (rs-fMRI) scans, self-supervised learning allows models to exploit the inherent structure and relationships present in the data, enabling the extraction of relevant features without the need for explicit labels. By formulating auxiliary tasks or pretext tasks, self-supervised learning models can effectively capture the underlying patterns and dependencies in the rs-fMRI data. These auxiliary tasks are designed to be easier to solve than the main task of interest, acting as a form of unsupervised training.

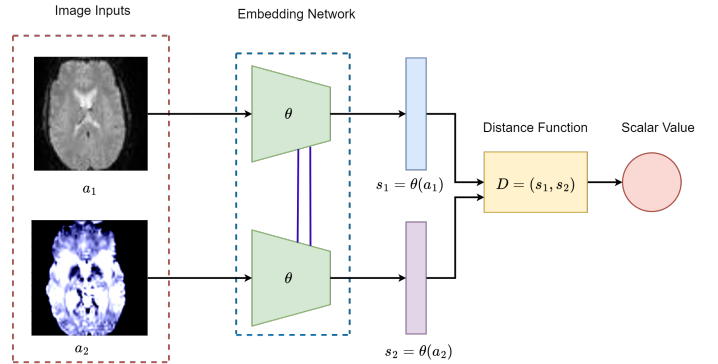


Fig. 2: Self-Supervised Learning Framework.

B. Contrastive learning and Non-Contrastive learning

One popular method for contrastive feature learning is SimCLR (Simple Framework for Contrastive Learning of Representations). SimCLR applies data augmentation techniques to generate diverse views of the rs-fMRI data. These augmented views are then embedded into a latent space where the model learns to maximize agreement between positive pairs and minimize agreement between negative pairs using contrastive loss functions. This process results in representations that capture essential information specific to each individual's neuroimaging data, thereby enhancing the accuracy of ASD diagnosis.

Non-contrastive feature learning methods, such as autoencoders, aim to reconstruct the input data by learning a compressed representation in the bottleneck layer. This reconstruction task encourages the model to capture important information about the underlying patterns present in the rs-fMRI data. SwAV utilizes clustering and swapping techniques to learn semantically meaningful features. By encouraging the model to produce similar representations for augmented views of the same input, SwAV captures relevant features that contribute to accurate ASD diagnosis.

Both contrastive and non-contrastive feature learning methods have shown effectiveness in extracting informative representations from rs-fMRI data, enabling accurate classification of individuals with ASD and improving clinical outcomes for affected individuals. By leveraging self-supervised learning techniques, such as contrastive and non-contrastive feature learning, researchers can exploit the vast amount of unlabeled rs-fMRI data to enhance the accuracy of ASD diagnosis and potentially uncover new insights into the neurobiological mechanisms associated with ASD.

C. Dataset

The dataset utilized is obtained from the Autism Brain Imaging Exchange (ABIDE) initiative and consists of resting-state functional magnetic resonance imaging (rs-fMRI) data. The dataset used in this research includes 1112 individuals, whose ages range from 7 to 64 years, with a median age of 14.6 years. To create the dataset, phenotypical samples from the available datasets are utilized. These samples include relevant demographic and clinical information, allowing for the extraction of brain scans and the construction of image datasets for analysis. To account for potential biases and confounding factors, such as age, gender, and site effects, we implemented several preprocessing and analytical strategies. First, we used the ABIDE I dataset, which includes demographic and site-related information, allowing us to monitor and adjust for these factors. We applied data normalization techniques across all samples to reduce site-related intensity differences in rs-fMRI scans. Additionally, we included balanced sampling to ensure a representative distribution across age and gender groups, minimizing skewness in the training and validation sets.

D. Data Preprocessing

The dataset is divided into training and validation sets following an 80:20 ratio. A series of transformations are applied to the images to augment the training data and capture various variations and orientations present in the neuroimaging data. These transformations include random horizontal flips, random vertical flips, and random rotations. Several augmentation techniques are applied to enhance model robustness and representation quality. Random cropping and Gaussian blur proved most impactful, with cropping allowing the model to capture localized brain activity patterns, and blur reducing sensitivity to noise. Random rotation improved rotational invariance, supporting feature extraction from various orientations, while intensity normalization ensured consistent representation by balancing intensity variations. Together, these augmentations enabled the model to generalize well across diverse rs-fMRI data, contributing to improved accuracy in ASD classification. Firstly, the images are resized to a standardized dimension of 224 x 224 pixels. This resizing step ensures consistency in the input image size, enabling efficient model training and inference. By resizing the images to a fixed size, we ensure that the models can handle input images of the same dimensions consistently. Batch normalization and standardization techniques were optimized based on the high dimensionality and noise in rs-fMRI data. Batch normalization mitigates the internal covariance shift, stabilizing the learning process, which is crucial in complex, high-dimensional datasets. We performed image standardization by rescaling pixel intensities to zero mean and unit variance, which reduces the influence of intensity variations across samples and enhances the discriminative power of learned features.

E. Contrastive learning with SimCLR

SimCLR (Simple Framework for Contrastive Learning of Representations) is utilized as this study's contrastive feature learning model. SimCLR [17] has demonstrated remarkable success in various domains and has gained attention for its ability to learn powerful representations from unlabeled data. Fig. 3 shows the SimCLR architecture for the autism classification task.

SimCLR consists of three key components: data augmentation, a base encoder network, and a contrastive loss function. Data augmentation techniques such as random cropping, color distortion, and Gaussian blur are applied to generate multiple augmented views of the same rs-fMRI scan. The base encoder network, often based on a deep convolutional neural network architecture such as ResNet, encodes the augmented views into a latent feature space. Algorithm 1 shows the pseudocode

Algorithm 1 SimCLR: Simple Framework for Contrastive Learning of Visual Representations

```

1: procedure SIMCLR( $N, s, f, T$ )
2:   Input: batch size  $N$ , constant  $s$ , structure of  $f$ , set of
      augmentations  $T$ 
3:   for sampled minibatch  $\{a_i\}_{i=1}^N$  do
4:     for all  $k \in \{1, \dots, N\}$  do
5:       Draw two aug functions  $t \sim T, t' \sim T$ 
6:       # Primary Augmentation
7:        $a_{2k-1} = t(a_k)$ 
8:       # Image Representation
9:        $r_{2k-1} = f(a_{2k-1})$ 
10:      # Image Projection
11:       $p_{2k-1} = g(r_{2k-1})$ 
12:      # Secondary Augmentation
13:       $b_{2k} = t'(b_k)$ 
14:      # Image Representation
15:       $r_{ak} = f(p_{2k})$ 
16:      # Image Projection
17:       $p_{2k} = g(r_{ak})$ 
18:    end for
19:    # pairwise similarity
20:    for all  $i \in \{1, \dots, 2N\}$  and  $j \in \{1, \dots, 2N\}$  do
21:       $x_{ij} = F(s_{ij})$ 
22:    end for
23:    Define  $E(i, j)$  as  $e(i, j) = -\log \frac{\exp(x_{ij})}{\sum_{j'=1}^{2N} \exp(x_{ij'})}$ 
24:     $C = \sum_{a_k \in \{a_1, \dots, a_N\}} (E(2k-1, 2k) + E(2k, 2k-1))$ 
25:    Update networks  $f$  and  $g$  to minimize  $C$ 
26:  end for
27:  Return: encoder network  $f$ , and discard  $g$ 
28: end procedure

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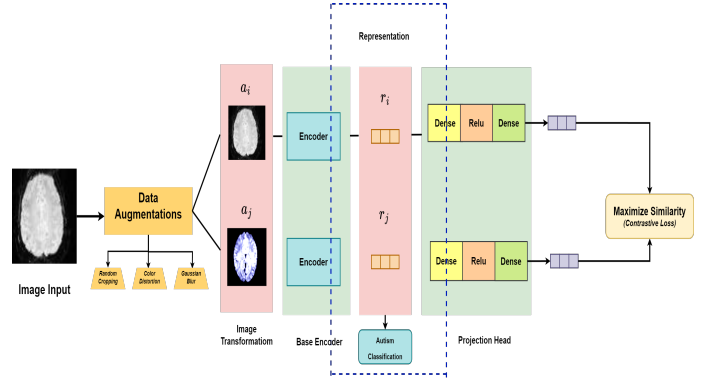


Fig. 3: SimCLR feature learning architecture.

for the SimCLR model. Augmentations were chosen based on their ability to capture the structural and functional variability in brain activity without distorting key features essential for ASD diagnosis. In neuroimaging, these augmentations aid in learning features that represent genuine patterns in brain connectivity, leading to improved representation quality. This approach allows the model to focus on invariant features critical for ASD classification.

The contrastive loss function employed by SimCLR is the InfoNCE (Normalized Mutual Information Estimation) loss. It maximizes agreement between positive pairs (augmented views from the same input) while minimizing agreement between negative pairs (augmented views from different inputs).

This encourages the model to learn discriminative features that capture the unique characteristics of an individual's rs-fMRI data. During training, the contrastive loss is computed by normalizing the similarity scores between pairs of augmented views using a temperature parameter. The model's parameters are optimized using backpropagation and gradient descent, updating the encoder network to improve the quality of the learned representations.

F. Non-Contrastive learning with SwAV

In addition to contrastive feature learning, non-contrastive feature learning methods are explored for ASD prediction. SwAV (Swapped Augmented Vectors) [18] is utilized as the non-contrastive feature learning model in this study. SwAV has shown effectiveness in capturing semantically meaningful features from unlabeled data. Fig. 4 shows the architecture for the SwAV framework.

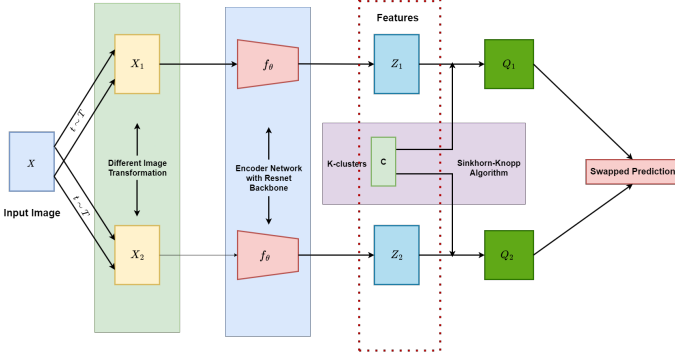


Fig. 4: SwAV feature learning architecture.

SwAV combines clustering and swapping techniques to encourage the model to produce similar representations for augmented views of the same input. By optimizing a clustering loss, the model learns to group augmented views from the same rs-fMRI scan together in the feature space. Additionally, a swapping loss is employed to ensure consistency between different views of the same input. Algorithm 2 shows the pseudocode for the SimCLR model. During training, SwAV optimizes these losses to learn representations that are invariant to data augmentations and preserve the underlying structure of the rs-fMRI data. The model is trained using backpropagation and gradient descent, iteratively updating the parameters of the encoder network to improve the learned representations.

One of the key components of SwAV is the Sinkhorn-Knopp algorithm, which is employed to facilitate the clustering process. In SwAV, the Sinkhorn-Knopp algorithm is employed to refine the transportation matrix that captures the relationships between the augmented views. The algorithm iteratively updates the matrix to balance the similarities between the augmented views, effectively aligning them in the latent space. This alignment encourages the model to produce similar representations for different views of the same input, enhancing the clustering and swapping objectives.

IV. RESULTS AND DISCUSSION

The study is performed on the machine with 16.00GB RAM and T4 GPU. Hyperparameters for this study used are 20 epochs, batch size of 32, learning rate of 0.001. For SimCLR temperature parameter of 0.5 is chosen whereas for the SwAV 3000 clusters is chosen for sinkhorn-knopp algorithm. Further for finetuning for ASD diagnosis in SimCLR and SwAV Resnet18 is used. The ResNet18 model is a deep convolutional neural network with 18 layers, specifically designed

Algorithm 2 SwAV: Swapping Assignments between Views

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1: procedure SwAV( $N, K, f, T$ )
2:   Input: batch size  $N$ , number of clusters  $K$ , structure
     of  $f$ , set of augmentations  $T$ 
3:   Initialize clusters centroids  $\{\mu_k\}_{k=1}^K$  randomly
4:   for sampled minibatch  $\{x_i\}_{i=1}^N$  do
5:     for all  $j \in \{1, \dots, N\}$  do
6:       Draw augmentation function  $t_j \sim T$ 
7:        $v_j = f(t_j(x_j))$ 
8:     end for
9:     Normalize features  $v_j$  across the batch dimension
10:    Assign samples  $v_j$  to cluster centroids  $\mu_k$  using
     Sinkhorn-Knopp algorithm
11:  end for
12:  Compute cluster assignments for each sample based
     on the cluster centroids
13:  for sampled minibatch  $\{x_i\}_{i=1}^N$  do
14:    for all  $j \in \{1, \dots, N\}$  do
15:      Draw augmentation function  $t_j \sim T$ 
16:       $v_j = f(t_j(x_j))$ 
17:    end for
18:    Normalize features  $v_j$  across the batch dimension
19:    Compute prototype features for each cluster cen-
     troid based on the assigned samples' features
20:    Compute assignment probabilities using the dot
     product between normalized features  $v_j$  and prototype
     features
21:  end for
22:  Update cluster centroids  $\{\mu_k\}_{k=1}^K$  using the EMA
     (Exponential Moving Average) update rule
23: end procedure

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to handle complex image data by enabling effective gradient flow through residual connections. The architecture consists of a series of convolutional layers grouped into four main stages, each with two residual blocks. Each residual block includes two convolutional layers with 3x3 filters and a stride of 1, followed by batch normalization and ReLU activation functions. This design, which includes residual connections, alleviates vanishing gradient issues, allowing for more efficient training and better feature learning across high-dimensional rs-fMRI data for ASD classification.

A. Evaluation

To assess the performance of the SimCLR and SwAV models for ASD diagnosis using neuroimaging data the performance metrics has been summarized given in Table 1.

TABLE I: Evaluation Results of SimCLR and SwAV Models

Metric	SimCLR	SwAV	DarkASDNet [19]	SVM [20]
Accuracy (%)	97.6	95.4	94.7	80.0
Precision (%)	95.7	99.3	94.5	81.0
Recall (%)	99.2	96.46	92.5	82.0
F1 Score (%)	97.4	97.82	95.0	82.5

Both models were trained for 20 epochs. The evaluation of the SimCLR model on the ASD dataset yields an accuracy of 97.6%. The precision score is recorded as 95.7%, indicating the model's ability to correctly classify positive samples. The recall score is 99.2%, reflecting the model's effectiveness in detecting individuals with ASD. The F1 score is calculated as 97.4%, representing a balanced measure of precision and recall. The results for both contrastive and non-contrastive models along with comparisons with deep learning and machine learning models are shown in Table I. Similarly,

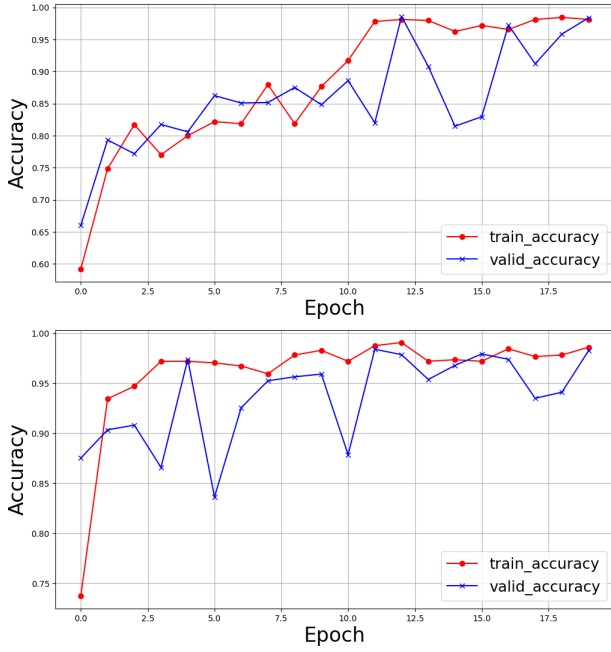


Fig. 5: Accuracy plots of training and validation for (i)SimCLR (ii)SwAV.

the evaluation of the SwAV model on the same ASD dataset yields an accuracy of 95.4%. The precision score is reported as 99.3%, indicating a high level of accuracy in identifying positive samples. The recall score is measured at 96.46%, demonstrating the model's proficiency in detecting individuals with ASD. The F1 score for the SwAV model is computed as 97.82%, reflecting a harmonious balance between precision and recall. These results highlight the strong performance of both the SimCLR and SwAV models in accurately diagnosing ASD from neuroimaging data. Fig 5 and Fig 6 represents the accuracy and loss plots for training and validation of SimCLR and SwAV model. respectively.

B. Experiments

In this section, we describe the experimental setup and performance comparison of the ResNet18, SimCLR, and SwAV models for ASD prediction. We conduct a series of experiments to evaluate the effectiveness of these models in capturing discriminative features from neuroimaging data. The backbone of the ResNet18 model is initially frozen for the first 10 epochs to allow the model to learn low-level features. After 10 epochs, the backbone is unfrozen, and the gradients are computed for backpropagation, enabling the model to fine-tune the learned features specific to the ASD prediction task. This sequential training strategy helps leverage the pre-trained knowledge of the backbone while adapting it to the specific task. Similarly, the SimCLR and SwAV models are trained with the same strategy, where the backbone is initially frozen and then fine-tuned for ASD prediction. This allows these models to learn meaningful representations from the unlabeled data and refine them to be more discriminative for the ASD classification task.

In this experiment we have investigated that when the SimCLR, SwAV, and Resnet18 models were frozen till 10 epochs the accuracy increased as seen in Fig 7. However as the models unfreeze after the 10th epoch the accuracy dropped significantly in the case of Resnet18 but the change in accuracy in SimCLR and SwAV models are not large and the accuracy increases afterwards. For the case of Resnet18, the accuracy

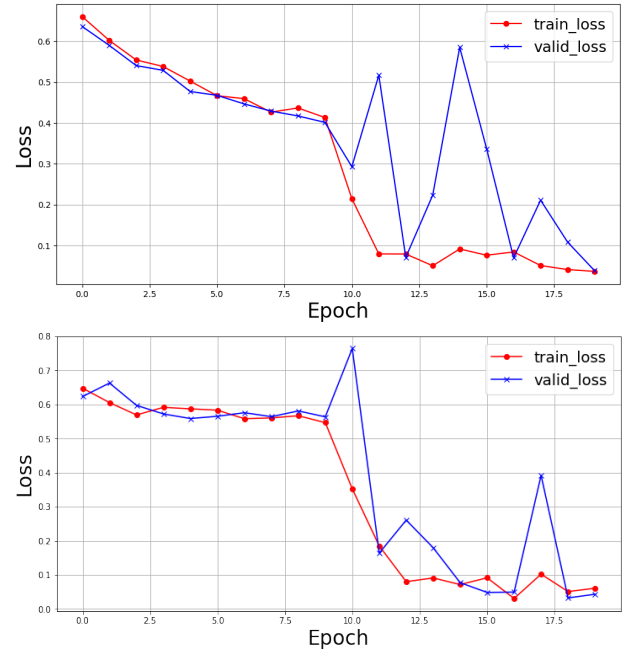


Fig. 6: Loss plots of training and validation for (i)SimCLR (ii)SwAV.

TABLE II: Results on frozen backbone till 10 epochs

Model	Accuracy (%)
SimCLR	98.3
SwAV	96.5
Resnet18	78.2

fluctuates afterwards and at the 20th epoch it achieved an accuracy of 78.2% while SimCLR recorded an accuracy of 98.8% and SwAV recorded an accuracy of 96.5% Similarly in the case of loss as shown in Fig 8. the loss decreases for all three models till the 10th epoch but after the 10th epoch there's a sudden increase in loss for all three models but the loss increases in SimCLR and SwAV models are quite low and on the other hand the loss significantly increased to a value of 42.3 and for the case of SimCLR loss peaked at 1.58 and for SwAV the loss peaked at 0.76. The results of the performance comparison between ResNet18, SimCLR, and SwAV models are presented in Table II. It can be observed

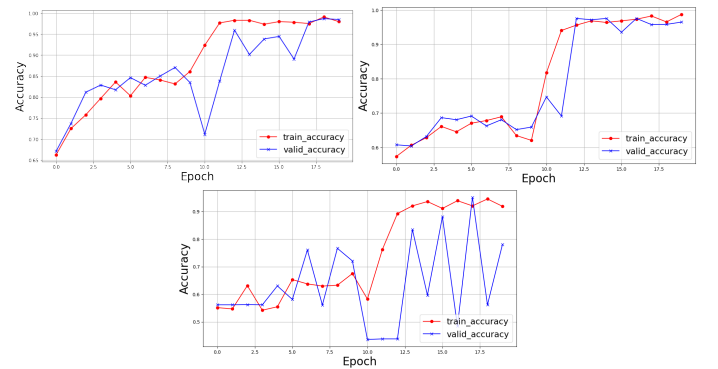


Fig. 7: Comparison of training and validation accuracy for (i) SimCLR (ii) SwAV (iii) Resnet18

that both SimCLR and SwAV outperform the ResNet18 model in terms of accuracy, precision, recall, and F1 score. T

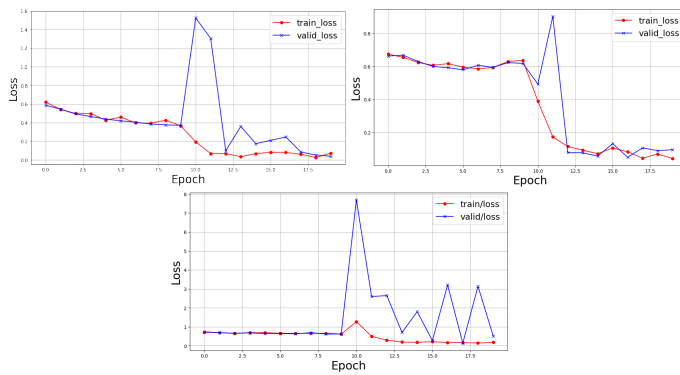


Fig. 8: Comparison of training and validation loss for (i) SimCLR (ii) SwAV (iii) Resnet18

C. Discussion

From Table I we can analyze that the SimCLR model achieved higher accuracy compared to SwAV due to its contrastive learning approach, which is highly effective in distinguishing subtle variations in high-dimensional rs-fMRI data. SimCLR utilizes a contrastive loss function to maximize agreement between different augmented views of the same sample, enabling it to capture more detailed, unique features from the ASD-related data. SwAV, being a non-contrastive method, focuses more on clustering and general representation, which may lack the fine-grained distinctions needed for this application. The temperature parameter in SimCLR's contrastive loss plays a crucial role in adjusting the focus on similar samples in high-dimensional space. Lower temperature values increase the sensitivity of the contrastive loss, making the model more discriminative by accentuating minor differences between classes. After experimentation, we set the temperature parameter to an optimal value of 0.5 that balances separation in the latent space while ensuring stable convergence, which is especially challenging in neuroimaging data where fine distinctions are essential. Further, Misclassifications, especially false negatives, were minimized using the contrastive loss in SimCLR, which strengthens the separation between ASD and non-ASD representations. For clinical applications like ASD diagnosis, reducing false negatives is critical. The model was further fine-tuned to focus on patterns specific to ASD, which helps mitigate the impact of misclassification on clinical diagnostics. By maximizing the contrastive loss margin, SimCLR reduces the chances of misidentifying ASD-positive samples.

V. CONCLUSION

In this study, we have employed two powerful pre-trained models, SimCLR and SwAV, to learn meaningful representations from the data in a pretext task setup. Our results demonstrate the effectiveness of SimCLR and SwAV models in capturing relevant information for the prediction of ASD. By leveraging self-supervised learning techniques on large-scale image datasets like ImageNet, these models are able to extract semantically meaningful features from the rs-fMRI data, even in the absence of labelled ground truth. This enables them to capture subtle variations and patterns that are indicative of ASD, enhancing the accuracy of the prediction task. The findings of this research have important implications for the field of ASD prediction and diagnosis. However, a limitation of this study is that it primarily relies on rs-fMRI data, which may not capture the full complexity of ASD-related

biomarkers. The use of pre-trained models trained on a pretext task provides a promising approach for leveraging large-scale unlabeled data to improve the accuracy and effectiveness of ASD prediction. These models offer a powerful tool for identifying potential markers and patterns associated with ASD, aiding in early detection and intervention strategies. Future research directions may involve exploring other self-supervised learning methods and incorporating multimodal data sources, such as genetic and behavioural data.

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